

Kambiz Tavabi

Data Scientist and Machine Learning Engineer | Business Intelligence Leader | PhD Neuroscientist

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PROFESSIONAL SUMMARY

Data Scientist with 15+ years building production-ready machine learning and NLP models—from pediatric neuroimaging biomarkers (AUC 0.86, published in Biological Psychiatry) to a live NLP classifier processing 110,000+ correctional incident records. Currently leads a 5-analyst team across cross-functional business areas at Washington State DOC, turning complex data into executive-ready dashboards and insights and automating reporting pipelines using SQL, Python, and Power BI. Applies AI and human-in-the-loop workflows to improve decision quality, and translates research methods into measurable policy and operational outcomes across government, clinical, and open-source domains as a PhD neuroscientist.

CORE COMPETENCIES

Multi-label NLP

Production ML

Power BI / SQL ETL

Statistical Modeling

MEG Neuroimaging

Open Source (MNE-BIDS)

Geospatial Analytics

Agile Product Leadership

WORK EXPERIENCE

Washington State Department of Corrections May 2023 – Present

Manager, Business Intelligence and Operations Surveillance (BIOS) | Research and Data Analytics Division – Tumwater, WA

- Lead a 5-analyst team across 5 cross-functional business areas (Prisons, Health Services, Reentry, Security, and executive leadership), delivering 20+ Power BI dashboards, ML prototypes, and applied research; partner with Recidiviz and CSG Justice Center on policy analytics
- Own a production multi-label NLP classifier for correctional-facility incident narratives: 73 labels across 110,928 training records, achieving 76% micro-averaged F1, 88% precision, and 30x improvement over a random baseline on 27,700 held-out incidents (scikit-learn TF-IDF with a multi-output one-vs-rest logistic regression model), deployed with human-in-the-loop review
- Ship 20+ Power BI dashboards and automated daily-to-annual reports (reception, dynamic risk-and-needs assessment, drug and alcohol screening, custody staffing, recidivism, behavioral misconduct, security-threat assessment, restrictive housing, and fentanyl task force) via SQL ETL, semantic models, and gateway refresh; led a SAS-to-SQL migration that eliminated recurring manual analyst intervention
- Built end-to-end Python analytics pipeline on 123,291 transport records (2019–2025, 39 counties): data-quality profiling, county-cooperation KPI scoring (0–100), and NetworkX transport-flow maps; resolved apparent 45% key-variable missing data to 0.8% structural gap via chi-square and Cramer's V ($V = 0.859$)
- Directed Medicaid all-payer-claims-database (APCD) linkage for the DOC release cohort (2016–2022): opioid prescribing, coverage, and post-release treatment outcomes in reproducible Quarto notebooks for population-health policy
- Established division-wide Power BI governance (quality-assurance standards, UX templates, permissions) and Git / Azure DevOps standards (README and CHANGELOG conventions, pull-request review), cutting onboarding time and standardizing delivery quality

Institute for Learning and Brain Sciences, University of Washington 2011 – 2023

Research Science Engineer – Seattle, WA

- Architected scalable Python magnetoencephalography (MEG) pipelines across a 12-year pediatric speech, reading, and language program; supported \$2M+ in NIH, NSF, and foundation-funded academic research with reproducible, standardized, neuroimaging data-structure-oriented workflows
- Co-developed an open-source Python package for structured data conversion and metadata tooling (reducing data-prep from hours to minutes; CI/CD on Windows, macOS, and Linux); deployed production MEG preprocessing for cohorts from children through adults
- Validated a combined noise-suppression method (~10 dB reduction in empty-room noise floor; single-trial localization $F(1,31) = 32.54$, $p < 0.001$), improving accuracy for dense sensor array MEG parameter guidance
- Delivered the first MEG source-modeling study of 14-month-olds during word processing: right frontal cortical activity predicted vocabulary growth ($\beta = 0.51$, $p = 0.001$); longitudinal brain-to-vocabulary correlation $r = 0.73$ ($p = 0.001$) at 27 months ($N = 22$, 5 follow-up assessments)
- Isolated a task-switching reading-skill biomarker in 42 children (ages 7–12): word-selective superior temporal gyrus (STG) responses correlated with reading skill at $r = 0.67$ ($p = 1e-6$) across two tasks, using statistical parametric mapping with spatiotemporal cluster-permutation testing

Post-Doctoral Researcher | Lurie Family Foundations MEG Imaging Center – Philadelphia, PA

- Validated MEG auditory discrimination neuromagnetic component latency as a language-impairment biomarker in autism spectrum disorders: AUC = 0.86, sensitivity 82%, specificity 71%, Cohen's $d = 3.11$ (~51 ms group delay) in 78 children (51 ASD, 27 controls), with blinded scoring and Bonferroni-corrected contrasts
- Built regulatory-minded analysis practices: a pre-specified bilateral STG source model, blinded brain response scoring, and linear mixed models across 301 observations for unbalanced pediatric repeated measures
- Characterized lexical oscillatory brain function signatures (theta-alpha desynchronization by hemisphere and lexicality; gamma-band synchronization near the word-uniqueness point) and semantic repetition priming (35% dipole attenuation) via beamforming source localization and time-frequency pipelines
- Delivered a 3-paper MEG program (Biological Psychiatry; NeuroReport, 2011a,b) spanning basic oscillatory neuroscience and translational pediatric diagnostics, under NIH grant R01-DC008871

SKILLS

Machine Learning and AI: scikit-learn, MultiOutputClassifier, multi-label classification, NLP pipelines, TF-IDF, k-fold and Jaccard evaluation, model validation, ROC/AUC analysis, experiment design, human-in-the-loop (HITL) deployment

Statistics: linear mixed models, growth-curve modeling, ANOVA, chi-square and Cramer's V association, effect sizes (Cohen's d), spatiotemporal cluster permutation testing, A/B and quasi-experimental design

Analytics and BI: Power BI (semantic models, DAX, gateway refresh), Power Automate, SQL / T-SQL / Oracle, ETL pipelines, SAS-to-SQL migration, Quarto / R Markdown reporting

Programming: Python (pandas, NumPy, SciPy), MATLAB, R, SQL

Geospatial and Networks: GeoPandas, NetworkX, U.S. Census TIGER/Line data, composite scoring

Neuroimaging and Scientific Python: MNE-Python, MNE-BIDS, MEG preprocessing, signal space separation, artifact rejection, Morlet wavelet time-frequency analysis, beamforming, dSPM, boundary-element modeling (BEM), FreeSurfer

Delivery and Infrastructure: Azure DevOps, Git, Agile product ownership, reproducible research workflows

PROJECTS

Correctional Incident NLP Classifier (IMRS)

Production multi-label classifier for 110,928 correctional-facility incident narratives (73 labels, 24-table export); MultiOutputClassifier one-vs-rest logistic regression with TF-IDF (SVM experimentation ~+5 F1); k-fold / Jaccard on 27,700 holdout – 75.6% micro-F1, 88.3% precision, 66.1% recall, 60.8% micro-Jaccard, 60.6% exact match (~30x random baseline); human-in-the-loop deployment at Washington State DOC.

Transport Analytics Pipeline

Python, GeoPandas, and NetworkX pipeline over 123,291 records (10 source tables to 32-field model, 39 counties); county-cooperation KPI, NetworkX flows, automated flagging of 4,107 high-complexity dual-mismatch individuals, Cramer's V missing-data audit ($V = 0.859$), and interactive Quarto HTML reports.

Long-Horizon Pediatric Cohort Data Engineering

12-year I-LABS MEG program infrastructure: scalable Python and MATLAB pipelines with signal space separation, artifact rejection, and multi-table integration under a \$2M+ NIH/NSF portfolio.

Auditory Oddball & Cortical Dynamics

Four-condition auditory oddball across two CHOP cohorts ($N = 96+$); repeated-measures ANOVA, Morlet wavelet time-frequency analysis, and beamforming; peer-reviewed lexical/spectral-temporal papers (NeuroReport).

Infant MEG Language Predictor

Longitudinal MEG source imaging with growth-curve models linking infant brain responses to later vocabulary ($r = 0.73$, $p = 0.001$); Developmental Cognitive Neuroscience, 2021.

Reading-Automaticity MEG Biomarker

Task-switching brain activation and reading-skill correlation ($r = 0.67$, $p = 1e-6$) in school-age children; Brain and Language, 2021.

ASD Language-Impairment Auditory Discrimination Biomarker

ROC-validated MEG latency classifier (AUC = 0.86, Cohen's $d = 3.11$); Biological Psychiatry, 2011.

EDUCATION

PhD, Neuroscience, University of Münster, Germany (2007)

MSc, Psychology, University of Oregon, USA (2004)

BSc, Physiology, University of California, Los Angeles, USA (2000)

PUBLICATIONS

Bosseler, A. N., Clarke, M., Tavabi, K., Larson, E. D., Hippe, D. S., Taulu, S., & Kuhl, P. K. (2021). Using magnetoencephalography to examine word recognition, lateralization, and future language skills in 14-month-old infants. *Developmental Cognitive Neuroscience*, 47, 100901. doi.org/10.1016/j.dcn.2020.100901

Bosseler, A., Clarke, M., Tavabi, K., & Kuhl, P. K. (2021). Using MEG to assess the neural mechanisms of phonetic distributional learning and future language growth in 2- and 6-month-old infants. *The Journal of the Acoustical Society of America*, 150(4), A111. doi.org/10.1121/10.0007797

Joo, S. J., Tavabi, K., Caffarra, S., & Yeatman, J. D. (2021). Automaticity in the reading circuitry. *Brain and Language*, 214, 104906. doi.org/10.1016/j.bandl.2020.104906

Clarke, M., Larson, E., Tavabi, K., & Taulu, S. (2020). Effectively combining temporal projection noise suppression methods in magnetoencephalography. *Journal of Neuroscience Methods*, 341, 108700. doi.org/10.1016/j.jneumeth.2020.108700

Appelhoff, S., Sanderson, M., Brooks, T., van Vliet, M., Quentin, R., Holdgraf, C., Chaumon, M., Mikulan, E., Tavabi, K., Höchenberger, R., Welke, D., Brunner, C., Rockhill, A., Larson, E., Gramfort, A., & Jas, M. (2019). MNE-BIDS: Organizing electrophysiological data into the BIDS format and facilitating their analysis. *Journal of Open Source Software*, 4(44), 1896. doi.org/10.21105/joss.01896

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Roberts, T. P. L., Cannon, K. M., Tavabi, K., Blaskey, L., Khan, S. Y., Monroe, J. F., Qasmieh, S., Levy, S. E., & Edgar, J. C. (2011). Auditory magnetic mismatch field latency: A biomarker for language impairment in autism. *Biological Psychiatry*, 70(3), 263–269. doi.org/10.1016/j.biopsych.2011.01.015

Tavabi, K., Embick, D., & Roberts, T. P. L. (2011a). Spectral-temporal analysis of cortical oscillations during lexical processing. *NeuroReport*, 22(10), 474–478. doi.org/10.1097/WNR.0b013e3283476b84

Tavabi, K., Embick, D., & Roberts, T. P. L. (2011b). Word repetition priming-induced oscillations in auditory cortex: A magnetoencephalography study. *NeuroReport*, 22(17), 887–891. doi.org/10.1097/WNR.0b013e32834ca576

Tavabi, K., Eling, L., Dobel, C., Pantev, C., & Zwitserlood, P. (2009). Effects of place of articulation changes on auditory neural activity: A magnetoencephalography study. *PLoS ONE*, 4(2), e4452. doi.org/10.1371/journal.pone.0004452

Tavabi, K., Obleser, J., Dobel, C., & Pantev, C. (2007). Auditory evoked fields differentially encode speech features: An MEG investigation of the P50m and N100m time courses during syllable processing. *European Journal of Neuroscience*, 25(10), 3155–3162. doi.org/10.1111/j.1460-9568.2007.05572.x

Villablanca, J. R., Schmanke, T. D., Crutcher, H. A., Sung, A. C., & Tavabi, K. (2000). The growth of the feline brain from fetal into adult life: II. A morphometric study of subcortical nuclei. *Developmental Brain Research*, 122(1), 21–33. [doi.org/10.1016/S0165-3806\(00\)00047-X](https://doi.org/10.1016/S0165-3806(00)00047-X)

AWARDS

- National Institutes of Health (NIH) Loan Repayment Award – Translational Research Excellence (2010)

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Introduction to Deep Learning and Neural Networks with Keras – IBM Deep Learning Professional Certificate, Coursera (2026). [Credential](#)
- Machine Learning with Python – IBM AI Engineering Professional Certificate, Coursera (2026). [Credential](#)
- Supervised Machine Learning: Regression and Classification – Coursera (2024). [Credential](#)
- Bayesian Statistics: From Concept to Data Analysis – Coursera (2023). [Credential](#)
- Data Visualization with Python – Coursera (2023). [Credential](#)
- Data Science Orientation – Coursera (2023). [Credential](#)
- Certified SQL Developer – W3Schools (2023). [Credential](#)
- Statistical Learning – edX (2022). [Credential](#)
- Elekta Neuromag MEG Advanced Program – Elekta, Helsinki, Finland (2012)

GRANT FUNDING

- Council of State Governments (CSG) Justice Center – Washington State DOC partnership
- National Institutes of Health (NIH): NINDS R01-NS104585; NICHD R21-HD092771 and R01-HD09586101; BRAIN Initiative / NIMH 1R24-MH114705; R01-DC008871 (PI: Timothy P. L. Roberts)
- National Science Foundation (NSF): BCS 1551330
- National Research Foundation of Korea: 2018K2A9A2A20088926 and 2019R1C1C1009383

- European Commission Horizon 2020: MSCA-IF-2018-837228-ENGRAVING; European Research Council Starting Grant SLAB ERC-YStG-676943; Academy of Finland 310988
- Foundations and other: Bezos Family Foundation; Simms Mann Foundation; Ready Mind Project Campaign (University of Washington); Washington State Life Sciences Discovery Fund; Google Summer of Code 2019; Nancy Lurie Marks Family Foundation; Pennsylvania Department of Health

OPEN SOURCE

- **MNE-BIDS** (co-developer) – Python package for organizing MEG, EEG, and intracranial-EEG data into the Brain Imaging Data Structure (BIDS); conversion routines, metadata extraction, anonymized-sharing and validator-oriented workflows, and multi-OS continuous integration for an international contributor community (Journal of Open Source Software, 2019; DOI 10.21105/joss.01896)
- **MNE-Python ecosystem** – Production preprocessing pipelines adopted across University of Washington pediatric neuroimaging studies